

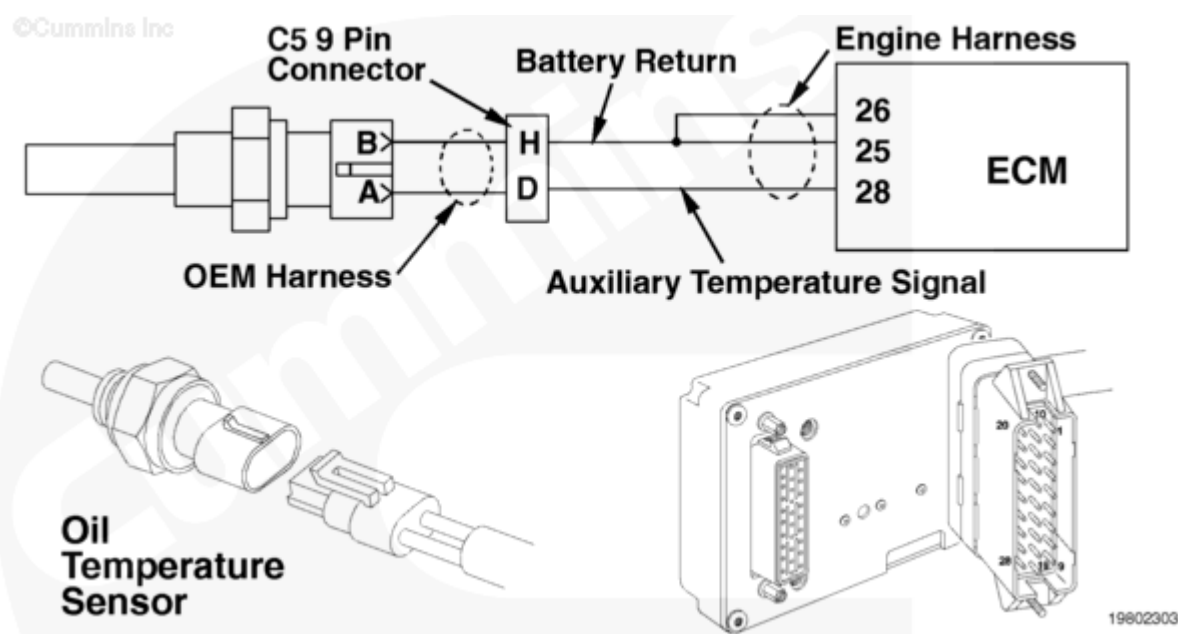
Fault Code: 521

Auxiliary Temperature Sensor Circuit - Voltage Above Normal or Shorted to High Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 521 PID(P): P177 SPN: FMI: 3 Lamp: On SRT: 00-645	High signal voltage detected at the auxiliary temperature sensor circuit.	None on performance.



Auxiliary Temperature Sensor Circuit

Circuit Description

The auxiliary temperature sensor is a variable resistor sensor used by the ECM to monitor the auxiliary temperature. The auxiliary temperature sensor has two circuits: signal, and return circuits. The signal voltage indicates the auxiliary temperature.

Component Location

The location of the auxiliary temperature sensor may vary and is OEM dependent.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected that the auxiliary temperature sensor signal voltage was out of range high.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.
- There is no engine protection system for auxiliary temperature on CENTRY™.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- There is no engine protection system for auxiliary temperature on CENTRY™.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Damaged OEM wiring harness.
- Damaged or loose connectors.
- Malfunctioning auxiliary temperature sensor.

Refer to Troubleshooting Fault Code t05-521 (</qs3/pubsys2/xml/en/procedures/98/98-t05-521.html>)

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